



Venus was habitable?

A theory suggests that a large ocean slowed down Venus' rotation, killing it. <https://digit.in/june19-48>



18 New Earth-Sized Exoplanets

Astronomers have found 18 Earth-sized exoplanets, including one that may be habitable. <https://digit.in/june19-49>

Gene Mapping

Gene mapping technologies are finally within the grasp of everyone since the costs of these tests are reducing dramatically since its inception.

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he origins of gene mapping date back to the 1910s and have developed into the intricate, sophisticated

medical process it is today after a series of scientific theories, technological developments and breakthroughs. Let's first find out what exactly they are.

Gene mapping is the process of sequentially allocating loci (fixed position of a gene on a chromosome) a relative position on a chromosome. In simple words, it finds the position of genes along a given chromosome. The distance between two genes is typically measured in map units (or recombination units). Gene maps comprise of genome markers

and/or genes and the distance of each marker from each other. The distances of the genome markers are calculated on the basis of the rate of occurrence of chromosome crossovers (recombinations) that occur during meiosis, and



Thomas Hunt Morgan discovered genetic linkage by studying fruit flies.

not by their physical location on the chromosome. Simply put, to calculate the distance or map units between two genes on a chromosome, we first need to calculate the recombination percentage of those two genes.

Gene mapping is done to map disease genes or trait loci which help in understanding the possible diseases a human being might develop and how to combat them.

DISCOVERY OF GENETIC LINKAGE AND RECOMBINATION

The discovery of the technique of genetic mapping happened in 1911 by Thomas Hunt Morgan, who chose a fruit fly to study genetics and genetic linking. After experimenting with fruit flies and several failed attempts of creating fruit flies that looked different from normal flies, Morgan discovered a fly that had white eyes (fruit flies usually have red eyes) which did not result from his experiments. After crossing the two types of flies together and witnessing several combinations of phenotypes, Morgan conceived the idea of genetic linkage, which basically means that when two genes are close to each other on a chromosome, they do not assort independently and are likely to be inherited together during meiosis. His experiments also suggested that genes were real, tangible objects that were present on chromosomes.

Morgan, along with his student Alfred Sturtevant, continued to conduct studies on genetic linkage which provided information about the positions of genes on chromosomes. This ultimately led to what we call gene mapping today.

While studying the chromosome theory of heredity, Morgan discovered that 'linked' traits of two chromosomes would sometimes separate. Meanwhile, other traits on the chromosome showed minimal linkage. This caused Morgan to propose recombination or crossovers. He said that two paired chromosomes could recombine or 'cross-over' to exchange information. Today, we know that this exchange happens during prophase of meiosis which creates multiple different combinations of alleles (alternate versions of genes) in the resulting